The librarian demo in MuJoCo: how contact force and deformation are simulated, what the skin panel plots, and when the skin helps

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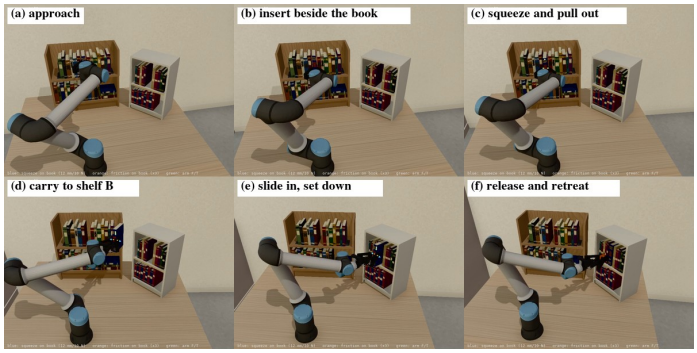


Figure 1: Key frames of the run (MuJoCo, Filament renderer). The blades slide into the 10 mm gaps beside the book (b); a 50 Hz controller squeezes, pulls the book out 21 cm and lifts it 8 cm (c); urx moves carry it to shelf B (d), slide it into the slot and set it down on a force threshold (e).

1 The demo, end to end

A UR10e with a Robotiq 2F-140 whose fingertips are our blades (`gripper_dohyun_v1`, a 0.5 mm PDMS layer on each flat face) takes a 40 mm book out of shelf A by approaching horizontally and shelves it in a 70 mm slot of shelf B, turned 45° toward the robot (Fig. 1). The task script uses only the `python-urx` calls of the real robot (`movel`, `speedl`, `get_tcp_force`, `gripper_action`, ...), so the same file drives the hardware. The video (`docs/media/librarian_demo.mp4`) has three synchronised panels: *left*, what the skin feels (Sec. 3); *centre*, MuJoCo rendered with the Filament physically based renderer plus a close-up that follows the tool, with arrows for the squeeze (blue) and friction (orange) on the book and the arm's wrench (green); *right*, the wrist camera and a urx console (joint angles, TCP pose and wrench, Robotiq registers, the last commands).

2 How MuJoCo produces force and displacement

2.1 Rigid bodies, joints and one step

The state is the generalised position q (joint angles, the pose of every free book, and the PDMS cells' displacements, Sec. 2.3) and velocity v. Each step of $h = 0.5$ ms solves

$$M(q)\,\dot{v} + c(q,v) = \tau + J(q)^{\top} f, \tag{1}$$

with M the mass matrix, c Coriolis/gravity, τ actuator and passive (spring, damper) forces, and f the *constraint forces*: contacts, the gripper's linkage (joint equalities) and joint limits, mapped by their Jacobian J [1]. Bodies are rigid: a book or a blade never strains. A **displacement** in MuJoCo is therefore a change of q — a body pose, a joint angle, or a degree of freedom we add on purpose to represent deformation (the PDMS layer).

2.2 Contact forces

Collision detection finds, for each pair of touching geoms, a contact frame (normal n, tangents t_1, t_2) and the signed distance r ($r < 0$: overlap). The contact force $f = (f_n, f_{t1}, f_{t2})$ must lie in the friction cone $\|f_t\| \le \mu f_n$, $f_n \ge 0$ (elliptic). MuJoCo computes all constraint forces of the step at once from a convex problem (Gauss's principle with *soft* constraints [2, 3]): the forces make each constraint's acceleration $a = J\dot{v}$ approach a reference

$$a^{\mathrm{ref}} = -b\,\dot{r} - k\,r, \tag{2}$$

a spring–damper set by `solref` (time constant, damping ratio), mixed with the unconstrained acceleration by an impedance $d \in (0,1)$ from `solimp` (regulariser $R = \frac{1-d}{d}\hat{A}$). Hence (i) contact forces are genuine forces in N: they balance gravity and actuation in Eq. (1) and obey action–reaction; (ii) contacts are slightly soft: at rest a small overlap r remains, about $\frac{1-d}{d}$ of the unresisted acceleration divided by k; (iii) friction is regularised: there is no static/kinetic difference and a sustained tangential load produces a slow creep (we keep it below 0.05 mm/s with a stiff friction-to-normal impedance ratio, `impratio` = 100; Sec. 6). Sensors are computed from the same solution: the UR10e's built-in wrist force/torque sensor is the wrench the flange transmits (MuJoCo `force`/`torque` sensors), minus the declared payload weight, as UR reports it.

2.3 The PDMS layer as a deformable body

MuJoCo has two ways to model deformable material [3]: *flex* elements (vertices are point masses; triangles or tetrahedra carry a Saint Venant–Kirchhoff finite-element elasticity with Young's modulus E and Poisson's ratio ν), and *lumped* models (rigid bodies joined by elastic joints or tendons). Our layer is $t = 0.5$ mm thick over 51×71 mm. We first built it as a 3D flex ($6 \times 8 \times 2$ vertices): on a fixed plate its shear stiffness was within 17% of the analytical value, but bolted to the moving finger its thin elements (aspect ratio 20:1 and more) made MuJoCo's effective-inertia solve ill-conditioned (contact and elastic forces no longer balanced). We therefore use the *thin-layer limit* of linear elasticity, which is exact when t is much smaller than the loaded area [4, 5]: each column of PDMS deforms under its own traction only (a Winkler foundation). The layer is split into 5×7 cells of area $A = 10.2 \times 10.1\ \mathrm{mm}^2$; each cell is a point mass bonded to the blade by three slide joints (compression u_z, shear u_x, u_y) with

$$k_n = \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}\frac{A}{t}, \qquad k_s = \frac{E}{2(1+\nu)}\frac{A}{t}, \tag{3}$$

(confined compression and shear of the column) and Kelvin–Voigt damping $\tau_d k$ (Fig. 2). With the placeholders $E = 1$ MPa (Sylgard-184-like [6]), $\nu = 0.45$, $\tau_d = 1$ ms: $k_n = 7.8 \times 10^5$ and $k_s = 7.1 \times 10^4$ N/m per cell. A 50 mg cell on such a spring oscillates at $\omega \approx 1.3 \times 10^5$ rad/s, so an explicit integrator would need $h < 2/\omega \approx 16\,\mu$s. MuJoCo's `discrete` integrator is implicit in position: it folds the stiffness into the effective inertia $\hat{M} = M + hD + h^2K$ and is stable at 0.5 ms; the layer then responds quasi-statically (in $\sim$1 ms), which is all a grasp needs. Each cell touches the object at one point (a flat ellipsoid filling the cell); that contact's own softness stands for the object's surface (a book cover), not for the PDMS.

3 What the skin panel plots

Table 1 separates what the physics model computes from what we *assume* about a sensor that is not characterised yet. The panel

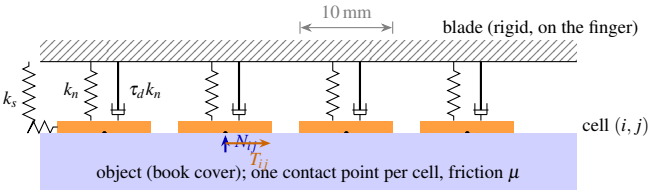


Figure 2: The PDMS layer as simulated: 5×7 columns of PDMS, each a point mass on three elastic slide joints (normal shown; two shear springs act along the face) bonded to the blade. Contact forces N_{ij}, T_{ij} come from MuJoCo's contact solver; the deformation u_{ij} is the cells' joint positions.

plotted quantity	how it is computed	status
N_{ij}, T_{ij} per cell	sum of the solver's contact forces on cell (i, j), in the PDMS frame	model physics
deformation u_{ij}	the cell's joint positions (Eq. 3)	placeholder E, ν
N, $\lvert T\rvert$, CoP, M	sums and moments over the cells	derived
margin q, Q	Eq. (4)	assumes $\mu = 1$
5×7 cells, 20 Hz	grid of the model; 50 ms frame averages	assumed sensor
noise, drift, hysteresis	none	not modelled
arm F/T	flange wrench − payload weight	as UR10e
slip, rotation	book pose relative to the TCP since grasp	sim only

Table 1: The skin panel, separated into what MuJoCo computes and what is assumed about the (uncharacterised) sensor. "Model physics": exact within the model, whose material values are placeholders.

shows, per blade, a map of the contact force per cell (colour: normal N_{ij}; arrows: shear T_{ij}), a map of the layer's deformation (colour: compression; arrows: shear displacement), and time series of $N = \sum N_{ij}$, $|T| = |\sum T_{ij}|$, the margin q with the Robotiq force setting, and the arm's wrench with the book's slip. In the quasi-static limit the cell's elastic force equals its contact force ($k_s u_{xy} = T_{ij}$, $k_n u_z = N_{ij}$, checked to within 2% in our tests), so the two maps carry the same information here; a real sensor may report either, or strain, through its own unknown transfer function. The margin of each blade treats its contact patch as a whole, using an elliptical limit surface for sliding and twisting [7, 8]:

$$q = 1 - \sqrt{\left(\frac{|T|}{\mu N}\right)^2 + \left(\frac{|M|}{\mu N \rho}\right)^2}, \quad \rho = \tfrac{2}{3}\sqrt{A_c/\pi}, \qquad (4)$$

where $M = \sum (r_{ij} - c) \times T_{ij}$ is the twist of the shear field about the centre of pressure c and A_c the loaded area (ρ of a uniformly loaded disc). $q = 1$: no load; $q \le 0$: the patch slides or twists. The grasp margin is the weaker blade, $Q = \min(q_L, q_R)$. Without the twist term the book rotated out of the grasp: it is held near its spine, 65 mm from its centre of mass.

4 Control, and who decides what

Arm. Every urx call becomes joint targets for MuJoCo position servos with gravity compensation (as the UR controller does): `movej` by trapezoidal joint profiles; `movel` by a straight TCP line (and an angle–axis rotation) converted every 2 ms with damped least-squares inverse kinematics $\Delta q = J^\top (JJ^\top + \lambda^2 I)^{-1} e$ [9]; `speedl` by integrating a TCP twist through the same IK. The UR10e's published DH parameters are used, so poses match the real controller. **Gripper.** Robotiq registers: position 0–255, speed, and force 0–255 mapped to 10–125 N per finger (2F-140 data sheet) as the finger actuator's torque limit. **Decisions.** No learning or LLM agent is in this loop: the phases and their switches are hand-written rules (Table 2), a placeholder for MPC/MPPI or a learned policy with the same Observation→Command interface. Only the contact phase uses the skin. On the real robot the script stops for a typed confirmation before any motion.

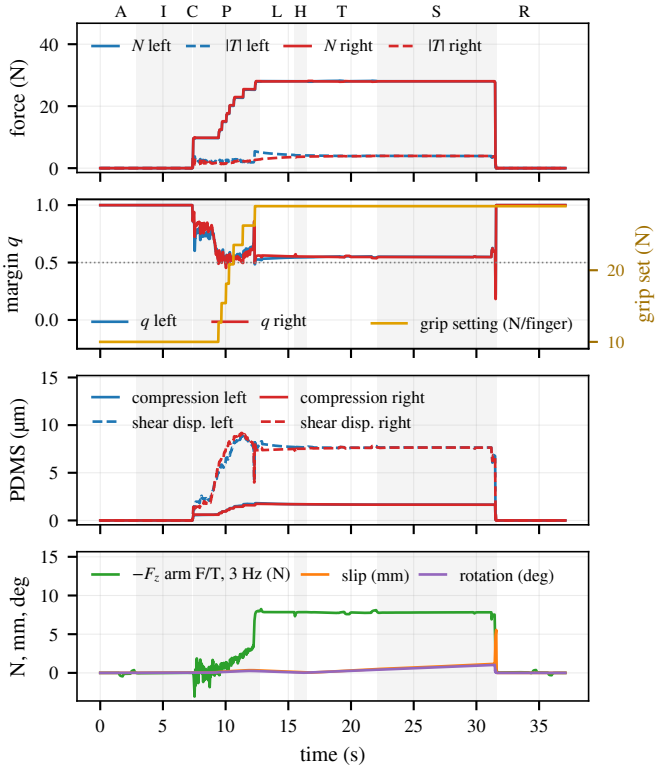


Figure 3: The demo run (0.8 kg book, $\mu = 1$, skin controller). Letters mark the phases: approach, insert, close, pull, lift, hold, carry (T), shelve (S), retreat. The controller raised the grip in steps until q settled at 0.55, just above $q^\star = 0.5$: 28 N per blade ($|T| = 3.9$ N). Holding, the PDMS compressed 1.7 μm and sheared up to 7.6 μm (a twist pattern: the book hangs 65 mm off the patch); the wrist F/T read the 7.8 N book. The book slipped 0.5 mm and 0.4° before set-down and landed 2 mm from the slot centre; neighbours moved 0.03 mm.

5 Ablation: when does the skin help?

Setup. The same task, 40 runs: five grip policies × book mass $\{0.4, 0.8, 1.6, 2.4\}$ kg × true PDMS–cover friction $\mu \in \{1.0, 0.5\}$. *skin*: the controller of Table 2, which always *assumes* $\hat{\mu} = 1$; *fixed*: the same motions at a constant force setting (10, 40, 70 or 125 N per finger) and no skin. A run succeeds if the book ends in shelf B's slot (centre within 15 mm, tilt and yaw within 5°). Successful runs placed the book within 7.7 mm, slipped $\le$1.7 mm and rotated $\le$1.4° in the grasp, and moved other books by $\le$0.03 mm.

Results (Table 3, Fig. 4). *(i) When the load is not known in advance, the skin replaces a worst-case squeeze with a per-book one.* With the right μ, it shelved all four books with 15, 28, 54 and 81 N per blade; to match that without the skin a fixed grip must be tuned for the heaviest book (70–125 N on every book), and grips tuned lower (10, 40 N) dropped or rotated the heavier books. *(ii) It does not help — it hurts — when μ is overestimated.* With $\mu = 0.5$ the skin policy failed on every book, while a 125 N grip succeeded on all. This is structural: during sliding $|T| = \mu N$, so the computed margin $q = 1 - |T|/(\hat{\mu} N) = 1 - \mu/\hat{\mu}$ stays at $\ge q^\star = 0.5$ whenever $\mu \le \hat{\mu}(1 - q^\star)$, and the force is never raised; the abort never fires either. A force-ratio margin is only as good as its μ. The skin will help here only with slip detection from the shear field (incipient slip starts at the patch rim [4, 11, 10]) or an online estimate of μ: the next step for the controller, and a requirement (resolution, rate) for the sensor's characterisation.

phase	signal	rule (who)
approach	—	`movel` 10 cm before the spines (script)
insert	wrist F/T	`movel` 12 cm, stop if $\|F\| > 15$ N (script)
close	skin N	close at 10 N/finger until both $N > 3$ N (controller)
pull, lift	skin Q	force setting $+2.7$ N per frame while $Q < 0.5$; arm pauses while $Q < 0.2$; abort if $Q < 0.2$ at full force for 1 s (controller)
carry	—	`movel` via a waypoint (script)
shelve	wrist F/T	stop on an unexpected 10 N; set down when the board pushes back 3 N; open (script)

Table 2: Phases, the signal that ends each one, and the rule. Thresholds are hand-tuned.

grip policy	shelved (of 4 masses) $\mu = 1.0$	$\mu = 0.5$	squeeze holding (N per blade)
skin, $\hat{\mu} = 1$	4	0	15 / 28 / 54 / 81
fixed 10 N	1	0	10
fixed 40 N	3	2	39
fixed 70 N	4	3	68
fixed 125 N	4	4	121

Table 3: Ablation summary (book mass 0.4 / 0.8 / 1.6 / 2.4 kg).

6 Limitations

- **The sensor is not modelled.** The panel shows what the model computes, on its 5×7 grid, at an assumed 20 Hz, without noise, drift or hysteresis. Calibrating the real skin's output against these quantities is open work, as are its resolution and rate.
- **Material values are placeholders.** $E = 1$ MPa, $\nu = 0.45$ (real PDMS: ≈0.49, which makes the confined layer several times stiffer in compression and leaves shear unchanged), $\tau_d = 1$ ms, $\mu = 1$; all are arguments of `make_blade_assets.py`. Deformations are micrometres either way (Fig. 3).
- **Friction** is Coulomb with one μ, no static/kinetic difference, and residual creep (≈0.02–0.05 mm/s under the book's twist) after raising MuJoCo's `impratio` to 100. Load sharing among cells depends on the contact softness, so read the maps qualitatively; net forces are exact.
- **The task is scripted**: hand-tuned thresholds, known shelf and book poses (no perception), one book geometry, and every book rigid (its cover compliance is the contact softness).

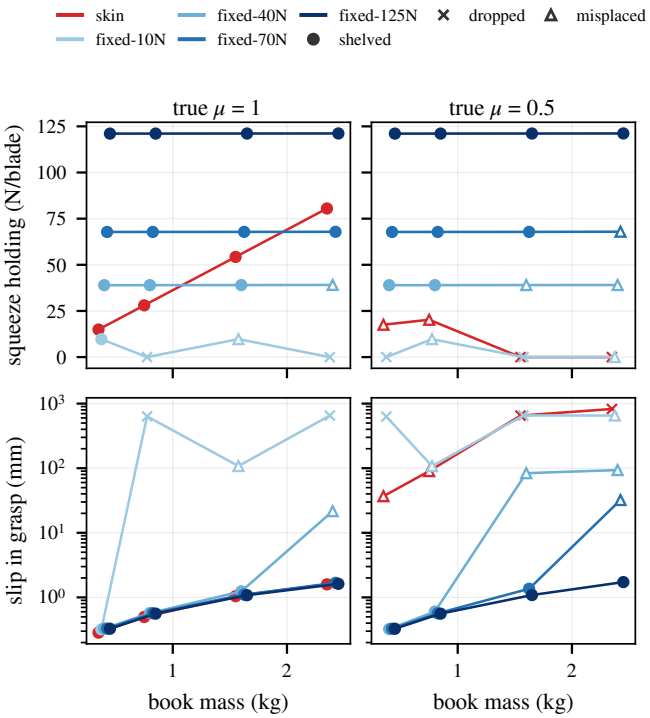


Figure 4: Ablation, 40 runs: squeeze while holding (top) and the book's slip in the grasp until set-down (bottom) against book mass, for the true friction $\mu = 1$ (left) and $\mu = 0.5$ (right). Filled markers: shelved; open: misplaced (△), dropped (×). The skin controller (red) assumes $\hat{\mu} = 1$.

References

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